MANAGEMENT OPERATING REVIEW - RECORD 020

Helioscope Robotics, Inc. - diligence file reference OPERATING_REVIEWS-020

Quarterly Business Review for Q3 2025: purpose and attendance

Pre-read materials went out three business days ahead, and the review started from questions rather than from a walkthrough. The Quarterly Business Review convened for Q3 2025. The next session is set for 2025-11-13. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. One function submitted its pre-read late. Notes were circulated on 2025-10-22. Requisitions open 17, Lucia Mbeki-Hart holding the list. Uptime 98.2 percent, interventions every 405 hours. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Headcount 247 of 259, Sunil Achterberg reporting.

One function submitted its pre-read late, and the section was taken last rather than allowed to shorten the discussion of others. Lucia Mbeki-Hart attended as Controller. Notes were circulated on 2025-10-22. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. The next session is set for 2025-11-13. The Quarterly Business Review convened for Q3 2025. Slipped deals 16, including Pinebank Distribution. Tobias Nkemdirim owns the Lidar assemblies recovery. Headcount 247 of 259, Sunil Achterberg reporting. Burn 10120 thousand, runway 15 months.

Data definitions were attached to the packet for the first time, and two functions discovered they were counting differently. The next session is set for 2025-11-13. The Quarterly Business Review convened for Q3 2025. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Sunil Achterberg, Director of Field Service, presented the operations section. Notes were circulated on 2025-10-22. Capital spend 910 thousand this period. Queue depth 33 sites, Fort Dodge at the front. Scrap 17 thousand, Penrose Castings contributing. Headcount 247 of 259, Sunil Achterberg reporting.

Action item review opened the meeting rather than closing it, which changed the tone of the whole session. Notes were circulated on 2025-10-22. One function submitted its pre-read late. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Sunil Achterberg, Director of Field Service, presented the operations section. The next session is set for 2025-11-13. Capital spend 910 thousand this period. Requisitions open 17, Lucia Mbeki-Hart holding the list. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Work in process 39 units, mostly Lidar assemblies.

Pipeline and stage conversion

Stage conversion tells a cleaner story than the headline number does, because the drop is concentrated between technical validation and commercial close. Sunil Achterberg owns the re-qualification pass. Weighted coverage sits near 158 open pursuits. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. The Bexley Pharmaceutical Wholesale pursuit re-dated again. Slippage totalled 16 opportunities against 11 before. Requisitions open 17, Lucia Mbeki-Hart holding the list. Uptime 98.2 percent, interventions every 405 hours. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Headcount 247 of 259, Sunil Achterberg reporting.

Deal inspection surfaced a habit of re-dating rather than re-qualifying, and the review asked for a hard rule on how many times a close date can move. Conversion into commercial close remains the weak stage. Slippage totalled 16 opportunities against 11 before. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Deals leaving the period: 16, most still dated. Weighted coverage sits near 158 open pursuits. Slipped deals 16, including Pinebank Distribution. Tobias Nkemdirim owns the Lidar assemblies recovery. Headcount 247 of 259, Sunil Achterberg reporting. Burn 10120 thousand, runway 15 months.

Qualification notes on the top ten pursuits were read aloud, which took most of the meeting and was the most useful thing in it. Weighted coverage sits near 158 open pursuits. Two losses went to the same competitor. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Deals leaving the period: 16, most still dated. The Bexley Pharmaceutical Wholesale pursuit re-dated again. Capital spend 910 thousand this period. Queue depth 33 sites, Fort Dodge at the front. Scrap 17 thousand, Penrose Castings contributing. Headcount 247 of 259, Sunil Achterberg reporting.

Multi-year commitments are becoming the norm in this segment, which stabilises the site queue and complicates the revenue phasing. The Bexley Pharmaceutical Wholesale pursuit re-dated again. Two losses went to the same competitor. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Weighted coverage sits near 158 open pursuits. Slippage totalled 16 opportunities against 11 before. Capital spend 910 thousand this period. Requisitions open 17, Lucia Mbeki-Hart holding the list. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Work in process 39 units, mostly Lidar assemblies.

Late-stage legal review has become a recurring source of slip, and the general counsel asked for the pattern rather than the anecdotes. Conversion into commercial close remains the weak stage. Sunil Achterberg owns the re-qualification pass. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Deals leaving the period: 16, most still dated. Two losses went to the same competitor. Ryemoor Wire Harness quoted lead times moved again. Capital spend 910 thousand this period. Cedar Bluff carries 33 of the queued fleet. Requisitions open 17, Lucia Mbeki-Hart holding the list.

Slippage this period was not random: the deals that moved had a common feature, which is that the customer had no internal project owner. Weighted coverage sits near 158 open pursuits. Slippage totalled 16 opportunities against 11 before. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. The Bexley Pharmaceutical Wholesale pursuit re-dated again. Deals leaving the period: 16, most still dated. Attrition 9 percent annualised in Southwest. Utilisation 84 percent, Crew Foxtrot fully committed. The HS-400 pallet mover took most of the test cell time. Columbus depot cleared its repair backlog.

Spares depot performance by part class, Q3 2025

Depot	Part class	Part lines stocked	Fill rate %	Backorders	Repair backlog units
Reno depot	mechanical	190	98	11	17
Columbus depot	mechanical	185	93	2	7
Dallas depot	mechanical	152	100	7	21
Atlanta depot	mechanical	231	95	0	0
Toronto depot	mechanical	184	100	8	15
Vehicle stock	mechanical	207	91	3	5
Reno depot	electrical and sensing	230	99	10	19
Columbus depot	electrical and sensing	197	94	5	1
Dallas depot	electrical and sensing	276	100	12	11
Atlanta depot	electrical and sensing	271	92	1	0
Toronto depot	electrical and sensing	252	97	9	13
Vehicle stock	electrical and sensing	247	90	4	3

Site queue and go-live performance

Readiness surveys are now a hard gate, and two sites were held at the gate rather than allowed to start with known network gaps. Fort Dodge and Elk Grove both cleared acceptance testing. Lucia Mbeki-Hart sequenced the queue by readiness. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Go-live cycle: 41 days, from 42. The queue carries 33 sites, 31 a period ago. Requisitions open 17, Lucia Mbeki-Hart holding the list. Uptime 98.2 percent, interventions every 405 hours. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Headcount 247 of 259, Sunil Achterberg reporting.

Customer-side readiness remains the single largest determinant of go-live date, and it is the variable the company controls least. Contract to go-live averaged 41 days. Cedar Bluff was held at the readiness gate. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Queue depth reached 33 sites. The queue carries 33 sites, 31 a period ago. Slipped deals 16, including Pinebank Distribution. Tobias Nkemdirim owns the Lidar assemblies recovery. Headcount 247 of 259, Sunil Achterberg reporting. Burn 10120 thousand, runway 15 months.

Parallel deployment at two sites in the same region worked for the first time, which changes what the queue can absorb. Queue depth reached 33 sites. The queue carries 33 sites, 31 a period ago. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Cedar Bluff was held at the readiness gate. Fort Dodge and Elk Grove both cleared acceptance testing. Capital spend 910 thousand this period. Queue depth 33 sites, Fort Dodge at the front. Scrap 17 thousand, Penrose Castings contributing. Headcount 247 of 259, Sunil Achterberg reporting.

The queue holds three sites with no confirmed dock date from the customer, and the review declined to schedule crews against hope. Cedar Bluff was held at the readiness gate. Go-live cycle: 41 days, from 42. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Queue depth reached 33 sites. Fort Dodge and Elk Grove both cleared acceptance testing. Capital spend 910 thousand this period. Requisitions open 17, Lucia Mbeki-Hart holding the list. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Work in process 39 units, mostly Lidar assemblies.

Days from contract to go-live improved, but the improvement came from the front of the process rather than from anything the installation crews changed. Cedar Bluff was held at the readiness gate. Contract to go-live averaged 41 days. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Queue depth reached 33 sites. Fort Dodge and Elk Grove both cleared acceptance testing. Ryemoor Wire Harness quoted lead times moved again. Capital spend 910 thousand this period. Cedar Bluff carries 33 of the queued fleet. Requisitions open 17, Lucia Mbeki-Hart holding the list.

The deployment playbook was rewritten mid-period after a crew found that the network prerequisites section was three revisions out of date. Fort Dodge and Elk Grove both cleared acceptance testing. Cedar Bluff was held at the readiness gate. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Contract to go-live averaged 41 days. The queue carries 33 sites, 31 a period ago. Attrition 9 percent annualised in Southwest. Utilisation 84 percent, Crew Foxtrot fully committed. The HS-400 pallet mover took most of the test cell time. Columbus depot cleared its repair backlog.

Warranty claims by build vintage and line, Q3 2025

Build vintage	Product line	Fleet units	Claims in period	Warranty cost per unit USD	Trend
2023 build	HS-200 tote handler	49	11	246	improving
2024 first half build	HS-200 tote handler	78	3	224	flat
2024 second half build	HS-200 tote handler	95	13	290	worsening
2025 first half build	HS-200 tote handler	136	13	268	improving sharply
2025 second half build	HS-200 tote handler	159	22	202	first period measured
2026 build	HS-200 tote handler	179	15	312	improving
2023 build	HS-400 pallet mover	46	6	224	flat
2024 first half build	HS-400 pallet mover	81	3	246	worsening
2024 second half build	HS-400 pallet mover	104	13	268	improving sharply
2025 first half build	HS-400 pallet mover	121	10	202	first period measured
2025 second half build	HS-400 pallet mover	162	22	312	improving
2026 build	HS-400 pallet mover	176	18	290	flat

Crew loading and certification

Certification standards were tightened rather than relaxed under schedule pressure, and the review recorded that choice deliberately. Crew leads remain the gating resource. Installer utilisation ran at 84 percent. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Tobias Nkemdirim covered two regions this period. Utilisation was 84 percent against 85 percent. Requisitions open 17, Lucia Mbeki-Hart holding the list. Uptime 98.2 percent, interventions every 405 hours. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Headcount 247 of 259, Sunil Achterberg reporting.

Utilisation above ninety percent leaves nothing for the unplanned second visit, and the second visit happens on roughly one site in four. Certified crew count trails the site queue of 33. Tulsa needed a second crew visit. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Tobias Nkemdirim covered two regions this period. Travel days per site fell again. Slipped deals 16, including Pinebank Distribution. Tobias Nkemdirim owns the Lidar assemblies recovery. Headcount 247 of 259, Sunil Achterberg reporting. Burn 10120 thousand, runway 15 months.

Handover from crew to service is still informal, and two accounts have complained about who to call after go-live. Utilisation was 84 percent against 85 percent. Certified crew count trails the site queue of 33. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Installer utilisation ran at 84 percent. Travel days per

site fell again. Capital spend 910 thousand this period. Queue depth 33 sites, Fort Dodge at the front. Scrap 17 thousand, Penrose Castings contributing. Headcount 247 of 259, Sunil Achterberg reporting.

Contractor coverage buys schedule protection at a cost per site that the review agreed to carry for a bounded number of jobs. Installer utilisation ran at 84 percent. Crew leads remain the gating resource. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Utilisation was 84 percent against 85 percent. Travel days per site fell again. Capital spend 910 thousand this period. Requisitions open 17, Lucia Mbeki-Hart holding the list. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Work in process 39 units, mostly Lidar assemblies.

Regional crew hubs changed the utilisation picture more than any hiring decision made in the same period. Certified crew count trails the site queue of 33. Travel days per site fell again. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Utilisation was 84 percent against 85 percent. Tobias Nkemdirim covered two regions this period. Ryemoor Wire Harness quoted lead times moved again. Capital spend 910 thousand this period. Cedar Bluff carries 33 of the queued fleet. Requisitions open 17, Lucia Mbeki-Hart holding the list.

Weekend and holiday work is filling gaps that the capacity model does not show, which makes the model look better than reality. Utilisation was 84 percent against 85 percent. Crew leads remain the gating resource. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Travel days per site fell again. Installer utilisation ran at 84 percent. Attrition 9 percent annualised in Southwest. Utilisation 84 percent, Crew Foxtrot fully committed. The HS-400 pallet mover took most of the test cell time. Columbus depot cleared its repair backlog.

Plant performance

The plant met its unit number and missed its labour-hours-per-unit target, which the review treated as the more important miss. Attainment gap: 1 units short of plan. Plan was 96 units; the line delivered 95. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Work in process ended at 39 units. Overtime carried the closing two weeks. Requisitions open 17, Lucia Mbeki-Hart holding the list. Uptime 98.2 percent, interventions every 405 hours. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Headcount 247 of 259, Sunil Achterberg reporting.

Line output has been steady for three periods while the plan has grown, which is a planning problem dressed up as an execution problem. Plan was 96 units; the line delivered 95. Attainment gap: 1 units short of plan. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Dana Whitlock owns the line rebalance. Output reached 95 units, from 90. Slipped deals 16, including Pinebank Distribution. Tobias Nkemdirim owns the Lidar assemblies recovery. Headcount 247 of 259, Sunil Achterberg reporting. Burn 10120 thousand, runway 15 months.

The build plan was cut mid-period rather than allowed to fail quietly, and the cut was communicated to deployment the same week. Plan was 96 units; the line delivered 95. Output reached 95 units, from 90. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. The HS-400 pallet mover drove most of the changeover time. Work in process ended at 39 units. Capital spend 910 thousand this period. Queue depth 33 sites, Fort Dodge at the front. Scrap 17 thousand, Penrose Castings contributing. Headcount 247 of 259, Sunil Achterberg reporting.

Second-shift ramp is going more slowly than planned because training the new operators takes line time from the first shift. Output reached 95 units, from 90. Plan was 96 units; the line delivered 95. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Work in process ended at 39 units. Overtime carried the closing two weeks. Capital spend 910 thousand this period. Requisitions open 17, Lucia Mbeki-Hart holding the list. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Work in process 39 units, mostly Lidar assemblies.

Test-cell capacity will bind before assembly capacity does, and the capital request to fix that has already been deferred once. Work in process ended at 39 units. Plan was 96 units; the line delivered 95. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Overtime carried the closing two weeks. The HS-400 pallet mover drove most of the changeover time. Ryemoor Wire Harness quoted lead times moved again. Capital spend 910 thousand this period. Cedar Bluff carries 33 of the queued fleet. Requisitions open 17, Lucia Mbeki-Hart holding the list.

Work instructions were re-photographed for the revised sensor head, ending a period where operators worked from a printout with a pen annotation. Attainment gap: 1 units short of plan. Dana Whitlock owns the line rebalance. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Work in process ended at 39 units. The HS-400 pallet mover drove most of the changeover time. Attrition 9 percent annualised in Southwest. Utilisation 84 percent, Crew Foxtrot fully committed. The HS-400 pallet mover took most of the test cell time. Columbus depot cleared its repair backlog.

Headcount plan against actual by function and location, Q3 2025

Function	Location	Plan	Actual	Gap	Open requisitions
Engineering	headquarters and plant	32	10	34	6
Manufacturing	headquarters and plant	19	5	22	4
Deployment and field service	headquarters and plant	11	4	11	2
Commercial	headquarters and plant	7	46	0	1
Customer success	headquarters and plant	6	35	0	1
Supply chain	headquarters and plant	3	27	0	0
Finance and administration	headquarters and plant	49	21	45	0
People	headquarters and plant	35	11	39	3
Engineering	field and regional	18	5	17	5
Manufacturing	field and regional	11	3	11	3
Deployment and field service	field and regional	6	2	5	1
Commercial	field and regional	4	24	0	0
Customer success	field and regional	3	18	0	0
Supply chain	field and regional	1	13	0	0
Finance and administration	field and regional	27	10	22	0
People	field and regional	20	5	19	2

Manufacturing quality

Scrap dollars tell a different story from scrap units, because the units being scrapped are the expensive ones. Renata Voss signed the corrective action. First-pass yield finished at 95 percent. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Scrap totalled 17 thousand dollars. Scrap ran 17 thousand, from 24 thousand. Requisitions open 17, Lucia Mbeki-Hart holding the list. Uptime 98.2 percent, interventions every 405 hours. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Headcount 247 of 259, Sunil Achterberg reporting.

The corrective action closed with evidence rather than with an assertion, and the evidence was three weeks of clean sampling data. Tessellate Controls changed a process without notice. Renata Voss signed the corrective action. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. First-pass yield finished at 95 percent. Scrap ran 17 thousand, from 24 thousand. Slipped deals 16, including Pinebank Distribution. Tobias Nkemdirim owns the Lidar assemblies recovery. Headcount 247 of 259, Sunil Achterberg reporting. Burn 10120 thousand, runway 15 months.

Inspection sampling reverted to the standard plan with a documented trigger, so the next excursion will be caught earlier. Tessellate Controls changed a process without notice. Renata Voss signed the corrective action. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Yield was 95 percent against 91 percent. Scrap ran 17 thousand, from 24 thousand. Capital spend 910 thousand this period. Queue depth 33 sites, Fort Dodge at the front. Scrap 17 thousand, Penrose Castings contributing. Headcount 247 of 259, Sunil Achterberg reporting.

Quality engineering asked for one more headcount and got a re-prioritisation instead, which the review acknowledged as a real trade. Scrap totalled 17 thousand dollars. Scrap ran 17 thousand, from 24 thousand. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Tessellate Controls changed a process without notice. Renata Voss signed the corrective action. Capital spend 910 thousand this period. Requisitions open 17,

Lucia Mbeki-Hart holding the list. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Work in process 39 units, mostly Lidar assemblies.

Process capability studies were completed on the two operations that had never had one, and one of them failed. Scrap totalled 17 thousand dollars. Tessellate Controls changed a process without notice. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Scrap ran 17 thousand, from 24 thousand. Renata Voss signed the corrective action. Ryemoor Wire Harness quoted lead times moved again. Capital spend 910 thousand this period. Cedar Bluff carries 33 of the queued fleet. Requisitions open 17, Lucia Mbeki-Hart holding the list.

Containment held: nothing built in the affected window left the plant without a hundred percent inspection record attached. Scrap ran 17 thousand, from 24 thousand. Yield was 95 percent against 91 percent. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Tessellate Controls changed a process without notice. Scrap totalled 17 thousand dollars. Attrition 9 percent annualised in Southwest. Utilisation 84 percent, Crew Foxtrot fully committed. The HS-400 pallet mover took most of the test cell time. Columbus depot cleared its repair backlog.

The problems we are still working

Constraint under review: field engineering capacity, at the resolved with new metric stage as of this session. The crew now runs eleven certified engineers against a plan of twelve, and the review replaced the open-requisition count with certified-crew-weeks as the capacity measure. Queue depth 33 sites, Fort Dodge at the front. Dana Whitlock and Renata Voss disagreed on sequencing.

Sharing a rolling forecast with the top suppliers cost nothing and moved two commit dates within a fortnight. Expedite activity rose with the build plan of 96. Penrose Castings improved after a site visit. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Supplier on-time delivery came in at 95 percent. Ryemoor Wire Harness missed two consecutive commit dates. Days outstanding 42, Quillon Home Supply the largest balance. Work in process 39 units, mostly Lidar assemblies. Columbus depot cleared its repair backlog. Utilisation 84 percent, Crew Foxtrot fully committed.

Supplier on-time delivery deteriorated across the board rather than at a single vendor, which points at demand signal quality on our side. Penrose Castings improved after a site visit. Supplier on-time delivery came in at 95 percent. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Expedite activity rose with the build plan of 96. Coverage weeks on long-lead parts were extended. Queue depth 33 sites, Fort Dodge at the front. Attrition 9 percent annualised in Southwest. Scrap 17 thousand, Penrose Castings contributing. Burn 10120 thousand, runway 15 months.

Requisition ageing and decisions, Q3 2025

Requisition	Role	Days open	Stage	Recruiting owner	Decision this review
REQ-2040	Field engineer	25	closed without hire	external agency	close and re-scope
REQ-2042	Crew lead	58	sourcing	hiring manager sourced	convert contractor
REQ-2044	Controls engineer	39	phone screen	Renata Voss	continue
REQ-2046	Manufacturing technician	68	on-site loop	external agency	raise band
REQ-2048	Buyer and planner	49	offer out	hiring manager sourced	widen radius
REQ-2050	Solutions architect	78	offer declined	hiring manager sourced	pause
REQ-2052	Service dispatcher	29	on hold	internal recruiting	close and re-scope
REQ-2054	Reliability engineer	62	closed without hire	external agency	convert contractor
REQ-2056	Field engineer (second opening)	43	sourcing	hiring manager sourced	continue
REQ-2058	Crew lead (second opening)	72	phone screen	external agency	raise band
REQ-2060	Controls engineer (second opening)	37	on-site loop	Renata Voss	widen radius
REQ-2062	Manufacturing technician (second opening)	58	offer out	internal recruiting	pause
REQ-2064	Buyer and planner (second opening)	25	offer declined	hiring manager sourced	close and re-scope
REQ-2066	Solutions architect	58	on hold	hiring manager	convert contractor

Requisition	Role	Days open	Stage	Recruiting owner	Decision this review
	(second opening)			sourced	
REQ-2068	Service dispatcher (second opening)	39	closed without hire	external agency	continue
REQ-2070	Reliability engineer (second opening)	68	sourcing	internal recruiting	raise band
REQ-2072	Field engineer (backfill)	49	phone screen	Renata Voss	widen radius

Inbound supply position

The harness supplier improved after a site visit, and the improvement is real enough to move them off the watch list. Sunil Achterberg issued the first supplier scorecards. Penrose Castings improved after a site visit. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Coverage weeks on long-lead parts were extended. Expedite activity rose with the build plan of 96. Requisitions open 17, Lucia Mbeki-Hart holding the list. Uptime 98.2 percent, interventions every 405 hours. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Headcount 247 of 259, Sunil Achterberg reporting.

Sharing a rolling forecast with the top suppliers cost nothing and moved two commit dates within a fortnight. On-time delivery was 95 percent, from 92 percent. Sunil Achterberg issued the first supplier scorecards. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Penrose Castings improved after a site visit. Supplier on-time delivery came in at 95 percent. Slipped deals 16, including Pinebank Distribution. Tobias Nkemdirim owns the Lidar assemblies recovery. Headcount 247 of 259, Sunil Achterberg reporting. Burn 10120 thousand, runway 15 months.

Supplier on-time delivery deteriorated across the board rather than at a single vendor, which points at demand signal quality on our side. Expedite activity rose with the build plan of 96. On-time delivery was 95 percent, from 92 percent. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Ryemoor Wire Harness missed two consecutive commit dates. Sunil Achterberg issued the first supplier scorecards. Capital spend 910 thousand this period. Queue depth 33 sites, Fort Dodge at the front. Scrap 17 thousand, Penrose Castings contributing. Headcount 247 of 259, Sunil Achterberg reporting.

Second-source qualification is slow, unglamorous work that removes the only single point of failure left in the drive train. Supplier on-time delivery came in at 95 percent. Ryemoor Wire Harness missed two consecutive commit dates. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Expedite activity rose with the build plan of 96. Coverage weeks on long-lead parts were extended. Capital spend 910 thousand this period. Requisitions open 17, Lucia Mbeki-Hart holding the list. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Work in process 39 units, mostly Lidar assemblies.

Component cost movement was mild, and the review spent its time on availability instead because that is what is hurting. Sunil Achterberg issued the first supplier scorecards. Expedite activity rose with the build plan of 96. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Ryemoor Wire Harness missed two consecutive commit dates. Supplier on-time delivery came in at 95 percent. Ryemoor Wire Harness quoted lead times moved again. Capital spend 910 thousand this period. Cedar Bluff carries 33 of the queued fleet. Requisitions open 17, Lucia Mbeki-Hart holding the list.

Supplier scorecards were issued for the first time with delivery, quality and responsiveness weighted, and the ranking surprised nobody. Ryemoor Wire Harness missed two consecutive commit dates. Expedite activity rose with the build plan of 96. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Sunil Achterberg issued the first supplier scorecards. Coverage weeks on long-lead parts were extended. Attrition 9 percent annualised in Southwest. Utilisation 84 percent, Crew Foxtrot fully committed. The HS-400 pallet mover took most of the test cell time. Columbus depot cleared its repair backlog.

Inventory and work in process

Kit shortages are concentrated in low-value parts, which is the most annoying possible failure mode. Work in process closed at 39 units. Lucia Mbeki-Hart owns the disposition decision. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Floor space in staging is now the binding limit. Work in process

moved from 42 to 39 units. Requisitions open 17, Lucia Mbeki-Hart holding the list. Uptime 98.2 percent, interventions every 405 hours. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Headcount 247 of 259, Sunil Achterberg reporting.

Coverage weeks by commodity is the only inventory number this meeting has ever acted on, so the others were dropped from the packet. Kit shortages were concentrated in low-value parts. Work in process closed at 39 units. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Buffer stock on the drive unit was raised deliberately. Lucia Mbeki-Hart owns the disposition decision. Slipped deals 16, including Pinebank Distribution. Tobias Nkemdirim owns the Lidar assemblies recovery. Headcount 247 of 259, Sunil Achterberg reporting. Burn 10120 thousand, runway 15 months.

A stock count in the sensor cage found more usable material than the system said, which is welcome and still a control failure. Buffer stock on the drive unit was raised deliberately. Work in process moved from 42 to 39 units. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Kit shortages were concentrated in low-value parts. Excess exposure sits in the superseded sensor head. Capital spend 910 thousand this period. Queue depth 33 sites, Fort Dodge at the front. Scrap 17 thousand, Penrose Castings contributing. Headcount 247 of 259, Sunil Achterberg reporting.

Safety stock policy was set when lead times were shorter, and nobody has revisited the parameters since. Work in process closed at 39 units. Work in process moved from 42 to 39 units. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Lucia Mbeki-Hart owns the disposition decision. Excess exposure sits in the superseded sensor head. Capital spend 910 thousand this period. Requisitions open 17, Lucia Mbeki-Hart holding the list. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Work in process 39 units, mostly Lidar assemblies.

Buffer inventory on the drive unit is the deliberate cost of the lead-time problem, and the review sized it explicitly. Buffer stock on the drive unit was raised deliberately. Lucia Mbeki-Hart owns the disposition decision. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Excess exposure sits in the superseded sensor head. Work in process closed at 39 units. Ryemoor Wire Harness quoted lead times moved again. Capital spend 910 thousand this period. Cedar Bluff carries 33 of the queued fleet. Requisitions open 17, Lucia Mbeki-Hart holding the list.

Spare-part inventory and production inventory are competing for the same components, and production keeps winning by default. Floor space in staging is now the binding limit. Excess exposure sits in the superseded sensor head. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Work in process moved from 42 to 39 units. Buffer stock on the drive unit was raised deliberately. Attrition 9 percent annualised in Southwest. Utilisation 84 percent, Crew Foxtrot fully committed. The HS-400 pallet mover took most of the test cell time. Columbus depot cleared its repair backlog.

Leavers and backfill position, Q3 2025

Function	Leavers in period	Annualised attrition %	Primary stated reason	Backfill position
Engineering	3	15	performance	requisition open
Manufacturing	1	4	travel load	backfilled internally
Deployment and field service	4	12	compensation	not backfilling
Commercial	0	7	relocation	contractor covering
Customer success	3	14	career change outside the industry	deferred to next quarter
Supply chain	1	5	manager relationship	requisition open
Finance and administration	3	10	retirement	backfilled internally
People	0	3	performance	not backfilling
Engineering (contract)	2	11	travel load	contractor covering
Manufacturing (contract)	0	6	compensation	deferred to next quarter
Deployment and field service (contract)	2	13	relocation	requisition open
Commercial (contract)	1	8	career change outside the industry	backfilled internally
Customer success (contract)	3	15	manager relationship	not backfilling

Field reliability review

A single navigation parameter set improved intervention rates at every site it was applied to, which is now the standard configuration. Tobias Nkemdirim leads the joint remediation plan. Fleet uptime averaged 98.2 percent. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Fort Dodge remains the weakest site on interventions. Interventions spaced 405 hours apart on the fleet median. Requisitions open 17, Lucia Mbeki-Hart holding the list. Uptime 98.2 percent, interventions every 405 hours. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Headcount 247 of 259, Sunil Achterberg reporting.

Intervention causes split roughly evenly between the robot, the facility and the operator, and only one of those three is ours to fix. Interventions spaced 405 hours apart on the fleet median. Fort Dodge remains the weakest site on interventions. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Uptime reached 98.2 percent, from 97.7 percent. Mean time between interventions was 405 hours. Slipped deals 16, including Pinebank Distribution. Tobias Nkemdirim owns the Lidar assemblies recovery. Headcount 247 of 259, Sunil Achterberg reporting. Burn 10120 thousand, runway 15 months.

Software releases are now staged across sites rather than pushed at once, after a release regressed docking behaviour at two accounts. Mean time between interventions was 405 hours. Uptime reached 98.2 percent, from 97.7 percent. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Interventions spaced 405 hours apart on the fleet median. Fort Dodge remains the weakest site on interventions. Capital spend 910 thousand this period. Queue depth 33 sites, Fort Dodge at the front. Scrap 17 thousand, Penrose Castings contributing. Headcount 247 of 259, Sunil Achterberg reporting.

Remote diagnostics closed more incidents without a visit than in any prior period, which is where the cost curve bends. Uptime reached 98.2 percent, from 97.7 percent. Tobias Nkemdirim leads the joint remediation plan. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Mean time between interventions was 405 hours. Interventions spaced 405 hours apart on the fleet median. Capital spend 910 thousand this period. Requisitions open 17, Lucia Mbeki-Hart holding the list. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Work in process 39 units, mostly Lidar assemblies.

Reliability engineering split the top failure modes by build vintage and found that one of them is entirely a legacy population problem. Quillon Home Supply measures uptime on a stricter basis. Interventions spaced 405 hours apart on the fleet median. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Uptime reached 98.2 percent, from 97.7 percent. Fort Dodge remains the weakest site on interventions. Ryemoor Wire Harness quoted lead times moved again. Capital spend 910 thousand this period. Cedar Bluff carries 33 of the queued fleet. Requisitions open 17, Lucia Mbeki-Hart holding the list.

Fleet uptime across all live sites improved, and the improvement is broad rather than driven by one recovering account. Interventions spaced 405 hours apart on the fleet median. Tobias Nkemdirim leads the joint remediation plan. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Fleet uptime averaged 98.2 percent. Mean time between interventions was 405 hours. Attrition 9 percent annualised in Southwest. Utilisation 84 percent, Crew Foxtrot fully committed. The HS-400 pallet mover took most of the test cell time. Columbus depot cleared its repair backlog.

Burn by category, Q3 2025

Category	Period USD thousands	Prior period USD thousands	Movement %	Commentary
Engineering payroll	716	105	581.9	headcount ramp
Manufacturing payroll and overheads	404	134	201.5	spares buy for the drive unit
Deployment labour, employees	607	672	-9.7	contract crews on two sites
Deployment labour, contract crews	506	470	7.7	contract crews on two sites
Commercial payroll	506	336	50.6	contract crews on two sites
Marketing and events	404	235	71.9	second hub fit-out
Field service payroll	303	235	28.9	second hub fit-out
Spares and consumables	202	168	20.2	second hub fit-out
Facilities and rent	404	201	101.0	second hub fit-out

Category	Period USD thousands	Prior period USD thousands	Movement %	Commentary
Information technology and tooling	202	134	50.7	headcount ramp
Professional services	303	168	80.4	overtime in the closing weeks
Recruiting fees	404	134	201.5	contract crews on two sites
Travel and subsistence	1922	100	1822.0	spares buy for the drive unit
Insurance and compliance	1518	67	2165.7	second hub fit-out
Test and validation materials	910	134	579.1	overtime in the closing weeks
Capital expenditure	809	67	1107.5	contract crews on two sites

Depot fill and warranty

Return material flow is slow, and the returned parts that could be reworked are sitting in a corner of receiving. Fill rate ran 96 percent against 91 percent. Warranty cost per fleet unit is falling. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Depot fill rate was 96 percent. Return material flow remains slow. Requisitions open 17, Lucia Mbeki-Hart holding the list. Uptime 98.2 percent, interventions every 405 hours. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Headcount 247 of 259, Sunil Achterberg reporting.

Critical spares coverage for the drive unit was raised deliberately while the supply constraint was live. Elk Grove took the planned sensor head replacements. Return material flow remains slow. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Depot fill rate was 96 percent. Vehicle stock is masking the true fill rate. Slipped deals 16, including Pinebank Distribution. Tobias Nkemdirim owns the Lidar assemblies recovery. Headcount 247 of 259, Sunil Achterberg reporting. Burn 10120 thousand, runway 15 months.

Depot technicians closed the repair backlog for the first time in five periods, which frees vehicle stock. Vehicle stock is masking the true fill rate. Return material flow remains slow. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Elk Grove took the planned sensor head replacements. Depot fill rate was 96 percent. Capital spend 910 thousand this period. Queue depth 33 sites, Fort Dodge at the front. Scrap 17 thousand, Penrose Castings contributing. Headcount 247 of 259, Sunil Achterberg reporting.

Courier spend tells the story that the fill rate hides, because the parts are moving from the wrong place. Depot fill rate was 96 percent. Vehicle stock is masking the true fill rate. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Elk Grove took the planned sensor head replacements. Dana Whitlock approved the critical spares increase. Capital spend 910 thousand this period. Requisitions open 17, Lucia Mbeki-Hart holding the list. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Work in process 39 units, mostly Lidar assemblies.

Spares fill rate from the regional depots governs how fast a field incident closes, and it has been the constraint at two sites. Elk Grove took the planned sensor head replacements. Warranty cost per fleet unit is falling. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Return material flow remains slow. Vehicle stock is masking the true fill rate. Ryemoor Wire Harness quoted lead times moved again. Capital spend 910 thousand this period. Cedar Bluff carries 33 of the queued fleet. Requisitions open 17, Lucia Mbeki-Hart holding the list.

Field engineers are carrying more parts in vehicles than the policy allows, which hides the true fill rate. Dana Whitlock approved the critical spares increase. Depot fill rate was 96 percent. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Elk Grove took the planned sensor head replacements. Return material flow remains slow. Attrition 9 percent annualised in Southwest. Utilisation 84 percent, Crew Foxtrot fully committed. The HS-400 pallet mover took most of the test cell time. Columbus depot cleared its repair backlog.

Headcount and recruiting

Compensation bands for field engineering were refreshed after three offers were declined on base salary. Annualised attrition sits at 9 percent. Median time to fill was 49 days. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. The headcount gap is 12 roles. Time to fill improved from 51 to 49 days. Requisitions open 17, Lucia Mbeki-Hart holding the list. Uptime 98.2 percent, interventions every 405 hours. Bexley

Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Headcount 247 of 259, Sunil Achterberg reporting.

Onboarding time to productivity for field engineers is measured now, and the number is longer than anyone assumed. Annualised attrition sits at 9 percent. Requisitions fell from 22 to 17. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Open requisitions number 17. The headcount gap is 12 roles. Slipped deals 16, including Pinebank Distribution. Tobias Nkemdirim owns the Lidar assemblies recovery. Headcount 247 of 259, Sunil Achterberg reporting. Burn 10120 thousand, runway 15 months.

Contractor-to-employee conversion is the cheapest source of certified crew, and the process for it does not exist yet. Median time to fill was 49 days. Annualised attrition sits at 9 percent. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Time to fill improved from 51 to 49 days. Requisitions fell from 22 to 17. Capital spend 910 thousand this period. Queue depth 33 sites, Fort Dodge at the front. Scrap 17 thousand, Penrose Castings contributing. Headcount 247 of 259, Sunil Achterberg reporting.

Headcount against plan closed part of its gap, and the remaining gap is concentrated in the two functions that gate deployment capacity. Median time to fill was 49 days. Annualised attrition sits at 9 percent. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Open requisitions number 17. Time to fill improved from 51 to 49 days. Capital spend 910 thousand this period. Requisitions open 17, Lucia Mbeki-Hart holding the list. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Work in process 39 units, mostly Lidar assemblies.

Open requisitions were re-ranked against the operating plan, and four were closed rather than left to age quietly. Headcount stood at 247 against a plan of 259. Open requisitions number 17. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Requisitions fell from 22 to 17. Annualised attrition sits at 9 percent. Ryemoor Wire Harness quoted lead times moved again. Capital spend 910 thousand this period. Cedar Bluff carries 33 of the queued fleet. Requisitions open 17, Lucia Mbeki-Hart holding the list.

Manager span of control has grown past what the review considers sustainable in two functions. Time to fill improved from 51 to 49 days. The headcount gap is 12 roles. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Median time to fill was 49 days. Annualised attrition sits at 9 percent. Attrition 9 percent annualised in Southwest. Utilisation 84 percent, Crew Foxtrot fully committed. The HS-400 pallet mover took most of the test cell time. Columbus depot cleared its repair backlog.

Collections and days outstanding by account, Q3 2025

Account	Invoiced USD thousands	Received USD thousands	Days outstanding	Collection blocker
Alder & Vance Retail	1141	661	33	payables portal submission
Northstar Cold Chain	931	2352	54	none
Havering Logistics	724	1693	39	customer audit of milestones
Quillon Home Supply	2484	1505	60	credit note pending
Tenmile Freight	1966	1223	27	purchase order number mismatch
Bexley Pharmaceutical Wholesale	1656	1035	51	acceptance document unsigned
Marrowfield Grocers	1449	941	36	payables portal submission
Pinebank Distribution	570	330	57	none
Alder & Vance Retail - second site	465	1176	30	customer audit of milestones
Northstar Cold Chain - second site	362	846	45	credit note pending
Havering Logistics - second site	1242	752	24	purchase order number mismatch
Quillon Home Supply - second site	983	611	48	acceptance document unsigned
Tenmile Freight - second site	828	517	33	payables portal submission
Bexley Pharmaceutical Wholesale - second site	724	470	54	none
Marrowfield Grocers - second site	380	220	39	customer audit of milestones

Decisions, owners and dates recorded in Q3 2025

An item that had rolled twice was closed by decision rather than by completion, with the rationale written down. Three items carried forward into 2025-11-13. Two due dates were changed in the room. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Tobias Nkemdirim, VP Engineering, took the largest open item. One item closed with measured evidence. Requisitions open 17, Lucia Mbeki-Hart holding the list. Uptime 98.2 percent, interventions every 405 hours. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Headcount 247 of 259, Sunil Achterberg reporting.

Due dates were set against the next meeting rather than against a vague month end, which improved follow-through visibly. Owners were named and dates set against 2025-11-13. Two due dates were changed in the room. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Three items carried forward into 2025-11-13. The oldest item dates from two sessions back. Slipped deals 16, including Pinebank Distribution. Tobias Nkemdirim owns the Lidar assemblies recovery. Headcount 247 of 259, Sunil Achterberg reporting. Burn 10120 thousand, runway 15 months.

Two owners pushed back on dates they had not been consulted about, and the dates were changed in the room. Tobias Nkemdirim, VP Engineering, took the largest open item. Owners were named and dates set against 2025-11-13. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Three items carried forward into 2025-11-13. One item closed with measured evidence. Capital spend 910 thousand this period. Queue depth 33 sites, Fort Dodge at the front. Scrap 17 thousand, Penrose Castings contributing. Headcount 247 of 259, Sunil Achterberg reporting.

Follow-up between meetings will be handled in writing, with the chair reading the updates rather than convening a session. One item closed with measured evidence. Tobias Nkemdirim, VP Engineering, took the largest open item. Reliability definitions were argued to a conclusion. One definition dispute took forty minutes. Two due dates were changed in the room. Three items carried forward into 2025-11-13. Capital spend 910 thousand this period. Requisitions open 17, Lucia Mbeki-Hart holding the list. Bexley Pharmaceutical Wholesale asked for the HS-400 pallet mover configuration. Work in process 39 units, mostly Lidar assemblies.

Action register as it stood after the Q3 2025 session

Action	Accountable owner	First raised	Revised due date	State	Note
OR-1803 Stand up crew lead certification pipeline	Priya Ramanathan	2025-09-04	2025-11-11	Open, date revised	partial
OR-2020 Measure onboarding time to productivity for crews	Renata Voss	2025-10-13	2025-11-02	In progress	outstanding
OR-2023 Set a plan freeze window inside the build month	Emil Farkas	2025-10-09	2025-11-08	Closed	decision recorded
OR-2026 Move collections calls to the account teams	Lucia Mbeki-Hart	2025-10-05	2025-11-14	Open	outstanding
OR-2029 Reissue the deployment playbook network section	Priya Ramanathan	2025-10-01	2025-11-20	In progress	outstanding
OR-2032 Split the reliability backlog by build vintage	Tobias Nkemdirim	2025-09-27	2025-11-26	Closed	memo circulated
OR-2035 Size the buffer stock on long-lead commodities	Hollis Bramwell	2025-09-23	2025-12-02	Open	outstanding
OR-2038 Refresh compensation bands for field engineering	Renata Voss	2025-09-19	2025-12-08	In progress	outstanding
OR-2041 Cost the second-shift ramp against overtime	Marcus Peil	2025-09-15	2025-12-14	Closed	report filed
OR-2044 Draft deposit terms for multi-site	Ingrid Solberg	2025-09-11	2025-12-20	Open	outstanding

Action	Accountable owner	First raised	Revised due date	State	Note
commitments					
OR-2047 Complete process capability study on the bond operation	Emil Farkas	2025-09-07	2025-12-26	In progress	outstanding
OR-2050 Decide disposition of the superseded sensor head stock	Marcus Peil	2025-09-03	2026-01-01	Closed	memo circulated
OR-2053 Read the tooling ownership clauses on four part numbers	Hollis Bramwell	2025-08-30	2026-01-07	Open	outstanding
OR-2056 Stage software releases across sites rather than at once	Griffin Tasse	2025-08-26	2026-01-13	In progress	outstanding
OR-2059 Bring deal-level notes for the top ten pursuits	Marguerite Do	2025-08-22	2026-01-19	Closed	memo circulated
OR-2062 Rebuild the site cost model separating retrofit from greenfield	Marcus Peil	2025-08-18	2026-01-25	Open	outstanding